rewards exploitation.

This is a desideratum of a BayesOpt policy, as it expresses our preference for exploration when all else is equal (that is, when the predictive means are equal) but also for exploitation when all else is equal (that is, when uncertainties are equal). We will see this property again in Chapter 5 with another BayesOpt policy called *Upper Confidence Bound*.

Computationally, we can initialize EI as a BayesOpt policy object using code that is almost identical to the code for PoI:

```
policy = botorch.acquisition.analytic.ExpectedImprovement(
    model, best_f=train_y.max()
)
```

Now, let's rerun our entire BayesOpt loop with the EI policy, making sure that we are starting with the same intial data set. This generates Figure 4.17, which is to be compared to Figure 4.13 of PoI.

Here, we see that while EI still focuses on the local optimal region around 2 initially, the policy quickly explores other regions in the search space to look for larger improvements from the incumbent. At the fifth iteration, we see that we are now inspecting the region on the left. Finally, after spending all ten queries, EI has successfully identified the global optimum of the objective function, outperforming PoI in the previous section.

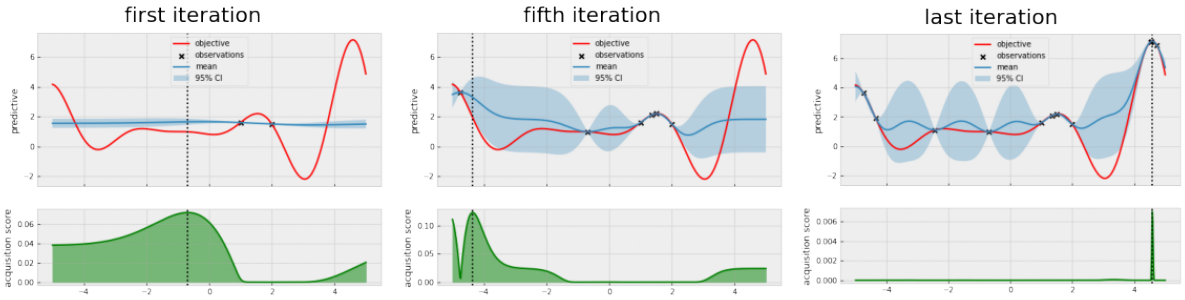


Figure 4.17 Progress made by the Expected Improvement policy. The policy balances exploration and exploitation better than Probability of Improvement and finds the global optimum at the end..

NOTE **Due to its simplicity and natural balance between exploration and exploitation, Expected Improvement is one of the most, if not the most, commonly used policies in Bayesian optimization. The policy is a good default option if there's no reason to prefer other policies.**

This marks the end of our chapter on improvement-based BayesOpt policies. It's important to keep the code that implements the BayesOpt loop for the Forrester function we used here in mind, as we will use it again to benchmark other policies in future chapters. Specifically, in Chapter 5, we will learn about BayesOpt policies that were inspired by the *multi-armed bandit* problem.

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